

Recipe book entries are no longer sorted in any meaningful manner

MC-123739 Crafting

Status	Resolved / Fixed
Affected	Minecraft 1.12.2, Minecraft 17w50a, Minecraft 1.13-pre1, Minecraft 1.13.1, 1.15.1, 1.15.2 Pre-release 2 (+22 more)
Fixed in	24w40a
Reported	2018-01-01

STEPS TO REPRODUCE

1. Create a Creative world with default settings.
2. Give the player all vanilla recipes by running `/recipe give @p *`.
3. Place down a crafting table.
4. Open the recipe book at the crafting table.
5. Set the recipe book to "Showing All" mode.
6. Observe that recipes within the categories are placed without a meaningful ordering, requiring the player to search through the entries to find a desired recipe.

OBSERVED BEHAVIOUR

Actual behavior

When the recipe book is opened at a crafting table with **Showing All** selected, recipe entries within each category are displayed in an arbitrary order rather than in a consistent, user-oriented order. Related recipes - such as recipes for the same block set or different color variants - are not grouped together, and entries are not arranged alphabetically or by another recognizable property.

The order can also change after reopening the recipe book or reloading the world, so the same recipe may appear in a different position between sessions. As a result, players must scan the available entries manually to find a recipe instead of being able to locate it predictably. This behavior occurs with the normal recipe-book collection after the known, fitting, search, and craftable filters are applied; the remaining entries are passed to the page layout without an apparent sorting step.

EXPECTED BEHAVIOUR

Recipe book entries should appear in a consistent, meaningful order within each category rather than being arranged randomly or changing after reloading. Recipes should be sorted in a predictable way, such as alphabetically, while related item and block sets remain grouped together, allowing players to locate desired recipes quickly and use the recipe book consistently.

ENVIRONMENT

Minecraft Java Edition Versions: 1.12.2, 17w50a, 1.13-pre1, 1.13.1, 1.15.1, 1.15.2 Pre-release 2, 1.15.2, 20w21a, 1.16.1, 20w27a, 21w03a, 1.16.5, 21w08b, 1.19.3, 1.20.2, 23w46a, 1.20.3 Pre-Release 1, 1.20.3 Pre-Release 3, 1.20.3 Pre-Release 4, 1.20.3, 1.20.4, 24w04a, 24w07a, 24w09a, 24w11a, 1.20.5 Pre-Release 1, 1.20.5, 1.21 (includes snapshots)

ORIGINAL DESCRIPTION

Recipe book entries are not sorted in any meaningful manner (even changing randomly upon reloading) leading to inconsistent and confusing usage of the recipe book. This is especially interesting, considering that the recipe book used to be sorted alphabetically, and that block sets were grouped together. Here is a demonstration:

2024-04-28_16-27-39.mp4

How to reproduce

- Give the player all vanilla recipes
`/recipe give @p *`
- Place down a crafting table
- Open the recipe book

- Set to "Showing All" mode

Observed & Expected behavior

- The item recipes will be randomly placed with no real rhyme or reason for where they are inside each category, leaving the player to search through each category randomly to find the item they want to craft.
- Each recipe entry would be neatly sorted by item type, colors, category, etc. just as the creative inventory is currently.

Suggested fix:

Either sort the recipe book categories by their alphabetical order, color, id, or just as the creative inventory is done.

Code analysis:

(Mappings: MCP Reborn 1.20.2 // Class: RecipeBookComponent.java // Method: updateCollections(boolean)):

```
private void updateCollections(boolean p_100383_) {
    List<RecipeCollection> list = this.book.getCollection(this.selectedTab.getCategory());
    list.forEach((p_296197_) -> p_296197_.canCraft(this.stackedContents, this.menu.getGridWidth(),
    this.menu.getGridHeight(), this.book));
    List<RecipeCollection> list1 = Lists.newArrayList(list);
    list1.removeIf((p_100368_) -> !p_100368_.hasKnownRecipes());
    list1.removeIf((p_100360_) -> !p_100360_.hasFitting());
    String s = this.searchBox.getValue();
    if (!s.isEmpty()) {
        ObjectSet<RecipeCollection> objectset = new ObjectLinkedOpenHashSet<>(this.minecraft.getSearchTree(Search
        Registry.RECIPE_COLLECTIONS).search(s.toLowerCase(Locale.ROOT)));
        list1.removeIf((p_301525_) -> !objectset.contains(p_301525_));
    }
    if (this.book.isFiltering(this.menu)) {
        list1.removeIf((p_100331_) -> !p_100331_.hasCraftable());
    }
    this.recipeBookPage.updateCollections(list1, p_100383_);
}
```

Here, the result list list1 is not sorted in any way before passing it to RecipeBookPage.updateCollections, which d

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-123739>